

Construction of an Affordable Wearable Apparatus for Severely Visually Impaired Pedestrians

Brian Liu¹, Michelle Luo¹, Maggie Wu¹, James Truglio¹

¹ Great Neck South High School, Great Neck, NY

Student Authors

Brian Liu* - High school

Michelle Luo* - High school

Maggie Wu* - High school

**These authors contributed equally to this work*

SUMMARY

Blind and visually impaired pedestrians frequently make unsafe decisions when navigating streets. Although solutions have risen to address this problem, they are often too costly. This study created an affordable wearable device that uses a YOLOv4-tiny model trained on a custom dataset to analyze the user's surroundings in real-time through a miniature camera, playing descriptive audio cues through earbuds (e.g. stop, go, countdown, move right, move left, crosswalk ahead). To evaluate the device, accuracy and performance metrics were recorded and usability survey results from legally blind testers were compiled. The model achieved a mean average precision (mAP) of 0.8734--similar to other studies applying YOLOv4-tiny. In the survey, participants indicated that the device was easily understandable, providing helpful cues (with relevant metrics rated 4 out of 5 or higher). However, its consistency (at 3 out of 5) was comparably lower, which may have led to users having less confidence in the device and anticipating to use it less frequently; this indicates that the device created in this study may be best used as a supplement to existing tools. Collectively, these results indicate that it is possible to create an affordable and user-friendly aid for severely visually impaired pedestrians.

INTRODUCTION

Around 2.2 billion people worldwide are blind or visually impaired (1). By 2030, an estimated 55 million people will suffer from total blindness and 330 million will suffer from moderate to severe visual impairment (2). These conditions are defined by visual acuities of individuals, where 6/6 constitutes "normal"--being able to see something that the average person can see from 6 meters away--and 6/12 (for example) indicates that someone would need to be 6 meters away to see something that the average person can see from 12 meters away. The International Classification of Diseases categorizes distance vision impairment in different levels: moderate corresponds to a visual acuity from 6/18 to 6/60, severe (legal blindness in the US) from 6/60 to 3/60, and blindness worse than 3/60 (1).

Severely visually impaired pedestrians make inconsistent and dangerous judgements when navigating streets, which jeopardizes their safety and may lead to physical harm and vehicular accidents. This is due to perceiving visual cues--such as distances between moving vehicles and traffic signals--with less clarity, resulting in unreliable decisions when crossing roads (3). Of 246 blind/low-vision participants in a study, 70 (28%) reported experiencing at least one collision with a vehicle in the past five years, while 158 (79%) reported having a near-collision (4). Many of these individuals took frequent walks: 77% of severely visually impaired participants and 66% of blind participants reported taking daily walks outdoors (4).

Another study found that 92% of blind participants and 90% of legally blind participants had tripped or fallen at least once, and that one-third of these falls resulted in medical ramifications (5). Suggested reasons for these falls include a general lack of awareness, unanticipated obstacles, and environmental misjudgements (5).

The most commonly used aids among visually impaired people include human guides, white canes, and guide dogs (6). However, each solution has its own drawbacks. Human guides have inconsistent availability and may restrict personal autonomy (7); guide dogs are very expensive to train and require regular care, with owners spending around \$30,000 on them (6); and white canes can only effectively help users detect lower obstacles and are typically used when users are already familiar with a given location (6). They also require the dedicated use of one hand, which may tire users; thus, a hands-free assistance device would be more practical (8). Moreover, current fixes for blindness itself, such as bionic eyes, stem cells, and gene therapy, have not been widely implemented, since they are risky for patients.

Additional solutions employ neural networks, which accurately and rapidly achieve computational tasks. One such task is object detection, where objects in an image are located and then classified. YOLOv4-tiny is an open-source lightweight object detection model that can be used in-real time environments and maintains a satisfactory balance between performance and accuracy relative to other models (9). YOLOv4-tiny is also computationally inexpensive, which lends itself useful to applications where portability is a concern, such as being run on a Raspberry Pi.

These navigation assistance systems can be put into four broad categories: hardware boards, smart canes, smart glasses, and smartphones (8). Frameworks implementing hardware boards (such as Raspberry Pis, laptops, or Arduinos) with external sensors (such as cameras or ultrasonic sensors) often struggle with a lack of portability; while smart canes can detect higher obstacles, they provide limited information about the user's greater surroundings; smart glasses, although portable and otherwise capable, are very expensive; and smartphone-based systems must often be positioned in a certain way or require an internet connection which could be inconvenient for the user (8). Since blindness is four times more common in less developed areas of the world, it is crucial to create a more affordable solution (1). Although very functional, Google Envision glasses cost \$3500 and biped.ai's obstacle avoidance system costs nearly \$2000. These have other functions besides street navigation, such as facial recognition, text-to-speech, and GPS. However, if cost, safety, and autonomy are primary considerations, it would be helpful to provide an affordable purpose-built device for blind pedestrians.

To this end, the goal of this study is to create an affordable, portable device to assist blind pedestrians in navigating streets. To do this, a YOLOv4-tiny model was trained on a custom crosswalk dataset, taken from a pedestrian's point of view. This was then transferred onto a Raspberry Pi 4 Model B, which connects to an external camera board (mounted to a pair of glasses), in-ear speakers, and power bank. Finally, the device was tested by legally blind pedestrians, who provided feedback via a survey.

RESULTS

To assess the accuracy of the YOLOv4-tiny model, mean average precision (mAP), precision, recall, and F1 scores were evaluated. mAP is a popular benchmark metric in object detection, summarizing the model's precision-recall curve across different object classes into a single value, where precision shows the reliability of the model's predictions and recall shows the predictions that the model should not have missed. Additionally, F1 score represents the harmonic mean of the precision and recall values, also combining them into a single value. After 10,000 iterations, the model had a mAP of 0.8734, precision of 0.79, recall of 0.88, and F1 score of 0.84 (Figures 1 and 2). Within the custom dataset, certain classes performed better than others: the car class had an mAP of 0.9286, the crosswalk class 0.9676, crosswalklightgo 0.8371, crosswalklightstop 0.7925, and sidewalkend 0.8372 (Table 2). mAPs were not obtained for crosswalklightblank and crosswalklightcountdown, as there were not enough images in the dataset.

To evaluate the usability of the device, legally blind individuals were invited to test it at quiet crosswalks with crossing lights while being monitored to ensure safety. After familiarizing themselves with the device, users filled out a questionnaire based on the System Usability Scale, where they rated nine parameters measuring effectiveness, efficiency, and satisfaction from 1-5 (10). Since device simplicity, ease of use, feature integration, intuitiveness, and feedback value averaged at least a 4.0, the device was easily understandable, providing useful cues (Figure 3 and Table 1). Following closely behind, comfort received a score of 3.9, showing that users had a relatively pleasant experience using the device (Figure 3 and Table 1). However, the consistency score was comparatively lower at 3.3, which could explain the low confidence that users had in the device and how they may not use it frequently (respectively indicated by scores of 2.9 and 3.5) (Figure 3 and Table 1).

DISCUSSION

The aim of this study was to create a wearable apparatus implementing a YOLOv4-tiny model trained with a custom dataset on a Raspberry Pi to assist pedestrians in real-time during street navigation. mAP was used as the primary metric to assess its accuracy. Additionally, to determine usability, participants filled out a survey after testing the device.

The device achieved a mAP of 0.8734, comparable to other studies using YOLOv4-tiny (Figure 1). It surpassed the mAPs of (11) and (12), which respectively utilized object detection of potholes on a Raspberry Pi for safety and features on thermal images for security, with mAPs of 0.80 and 0.78. Another study which employed YOLOv4-tiny to detect electronic components had a mAP of 0.94 for its model (9). The varying mAPs across studies could be attributed to the unique target objects being detected, each with differing levels of distinctiveness; for example, in (9), the camera was affixed to depict objects on a flat green background, rather than in fluctuating environments.

There was also variation in mAPs across different classes of the custom dataset. The car and crosswalk classes had the greatest mAPs (0.9286 and 0.9676), as they appeared frequently in training videos, having many annotations (Table 2). With a mAP of 0.8372, the sidewalk class did not perform as well—even with the second highest number of annotations. This could be attributed to variation in sidewalk end design; since they are not as homogenous as cars or crosswalks, they are more difficult for the model to reliably predict. Moreover, crosswalklightgo and crosswalklightstop likely had comparably low mAPs due to the camera quality of the Raspberry Pi camera module v2, which may result in predictions made from insufficient visual data (exacerbated by the small size of the symbols) and ultimately more false positives and false negatives (13).

The survey showed that the device was easy to understand and provided useful feedback to users while being reasonably comfortable (Figure 3). However, its consistency was lower, which likely led to users having less confidence in the device and expecting to use it less frequently (Figure 3). These results indicate that the device may be best used as a supplement to existing tools for the blind, such as white canes. Nonetheless, the survey was limited—it did not have a large sample size and all participants were legally blind (not fully blind), so its results may only be applied to a certain extent in the greater blind population. Thus, the device in its current state would be best used as a supplement to existing forms of aid such as white canes, rather than an independent form of assistance.

In future research, more variation in lighting conditions should be incorporated in the dataset to help the device work more reliably at different times of day. Also, since newer YOLO

models (such as YOLOv8-small) have been released this study began, comparing them will help to optimize the device. Additionally, wireless earbuds could be integrated into the device for convenience--however, this would result in having more components to charge. Finally, more computing power (in the form of an Intel Movidius Neural Compute Stick) could be implemented to address the small delay between the camera's observations and the model's feedback. Other computers, such as the Nvidia Jetson Nano, may also be tested. This would likely result in greater consistency, improving the other survey metrics and user satisfaction as a whole. However, it would result in a less affordable device; therefore, optimization of the model itself should be prioritized.

The apparatus created in this study offers an affordable and usable solution for severely visually-impaired pedestrians. It had a mAP of 0.8734, and surveyed participants found it helpful and easy to use. However, consistency was lacking--this project serves as a proof of concept, and with further refinement, it may be more widely implemented.

MATERIALS AND METHODS

Creating the custom dataset

To make the custom dataset for training the model, 1815 frames from around 100 manually recorded videos of Great Neck street crossings from a pedestrian's point of view were annotated on Roboflow (a web tool that assists with computer vision). Then, it was split into training and testing subsets, following an 80:20 ratio. For the videos in the training subset, bounding boxes were manually labeled with different classes, which include car, crosswalk, crosswalklightcountdown, crosswalklightgo, crosswalklightstop, and sidewalkend (Figure 4). Three additional open-source datasets from Roboflow, listed in the appendix, were used to boost the model's accuracy, increasing the total number of annotated frames to 2700. Before training the model with this dataset, it was preprocessed for the model to effectively interpret--resized to 680 x 460 pixels. Then, training data was augmented by means of varied rotation (-15 to 15 degrees) through Roboflow to prevent overfitting of the model on too many similar images and account for discrepancies in camera angles for real-world usage. In total, around 6700 images were exported from Roboflow.

Developing the YOLOv4-tiny model

The lightweight YOLOv4-tiny model, trained in Google Colaboratory (Colab), was used for object detection in this study. The dataset and Darknet GitHub repository were imported to

Colab, where the model trained itself, adjusting weights and biases to maximize its accuracy. The best weights were saved in a file to be used when testing. After training, the following metrics were obtained: mean average precision (mAP), precision, recall, F1-score, and confusion matrix (depicting true positives--TP, false positives--FP, and false negatives--FN) (Figure 1).

The precision of a model, which communicates the reliability of its positive predictions, is found using the equation $TP / (TP + FP)$ (Jiang et al., 2022). Recall, which depicts the negative predictions which should have been positive, is found using $TP / (TP + FN)$ (Jiang et al., 2022). mAP is a popular metric to evaluate the accuracy of object detection models, representing the area under the precision-recall curve, and F1 score represents the optimal balance of precision and recall values, combining them as a harmonic mean (14; 15).

Design and function of the smart hat

To implement the YOLOv4-tiny model in the RPi, the model was downloaded from Colab into the RPi's local storage system. Next, code was written to synthesize the model's predictions, using the RPi camera to capture video and give the user real-time instructions in street navigation. To do this, the pygame (audio) and cv2 (video and computer vision) libraries were used. First, the RPi greets the user once its startup process begins. While in use, it uses the sizes and locations of bounding boxes, as well as timers, to dictate the relevant audio cue. These inform the user of: approaching a sidewalk end, approaching a crosswalk, various signals on crosswalk lights (stop, go, countdown), cars stopping (when it is safe to cross the road), successfully crossing the road (upon reaching the opposite sidewalk end), and whether to move right or left to align themselves with a sidewalk end or crosswalk.

The smart hat consists of the Raspberry Pi camera module v2 mounted to the brim of a hat, connected to a Raspberry Pi 4 Model B (RPi) running on a power bank (Figure 5). Earbuds were used to relay auditory feedback describing the user's surrounding environment (Figure 5).

ACKNOWLEDGEMENTS

The authors are grateful for the resources provided by Great Neck South High School, which made this study possible.

REFERENCES

1. Liste, Sebastian. "Blindness and Vision Impairment." *World Health Organization*, 13 Oct. 2022. Accessed 13 June 2023.
2. Ackland, Peter, et al. "World Blindness and Visual Impairment: Despite Many Successes, the Problem Is Growing." *PubMed Central*, U.S. National Library of Medicine, 8 Feb. 2018.
3. Hassan, Shirin E. "Are Normally Sighted, Visually Impaired, and Blind Pedestrians Accurate and Reliable at Making Street Crossing Decisions?" *Investigative Ophthalmology & Visual Science*, The Association for Research in Vision and Ophthalmology, 1 May 2012.
4. Liu, Sara. "The Impact of Electric/Hybrid Vehicles and Bicycles on Pedestrians Who Are Blind or Have Low Vision." *Monash University*, Vision Australia, Oct. 2018.
5. Manduchi, Roberto, and Sri Kurniawan. "Mobility-Related Accidents Experienced by People with Visual Impairment." *Semantic Scholar*, University of Santa Cruz, 9 Feb. 2011.
6. Jeamwatthanachai, Watthanasak, et al. "Indoor Navigation by Blind People: Behaviors and Challenges in Unfamiliar Spaces and Buildings." *Sage Journals*, Sage, 8 May 2019.
7. Garaj, V, et al. "A System for Remote Sighted Guidance of Visually Impaired Pedestrians." *Sage Journals*, Sage, May 2003.
8. Kuriakose, Bineeth, et al. "DeepNAVI: A Deep Learning Based Smartphone Navigation Assistant for People with Visual Impairments." *ScienceDirect*, Elsevier, Feb. 2023.
9. Guo, Ce, et al. "Improved YOLO4-Tiny Network for Real-Time Electronic Component Detection." *Nature News*, Nature Publishing Group, 23 Nov. 2021.
10. Brooke, John. "SUS - A Quick and Dirty Usability Scale." *ResearchGate*, 1996.
11. Asad, Muhammad Haroon, et al. "Pothole Detection Using Deep Learning: A Real-Time and AI-on-the-Edge Perspective." *Advances in Civil Engineering*, Hindawi, 20 Apr. 2022.
12. Jiang, Chenchen, et al. "Object Detection from UAV Thermal Infrared Images and Videos Using YOLO Models." *International Journal of Applied Earth Observation and Geoinformation*, Elsevier, 18 July 2022.
13. Hao, Yu, et al. "Understanding the Impact of Image Quality and Distance of Objects to Object Detection Performance." *arXiv.Org*, 17 Sept. 2022.
14. Zhang¹, Dongyang, et al. "Smart-YOLO: A Light-Weight Real-Time Object Detection Network." *Journal of Physics: Conference Series*, IOP Publishing, 1 Jan. 2021.
15. Wang, Yan, et al. "DSE-Yolo: Detail Semantics Enhancement YOLO for Multi-Stage Strawberry Detection." *Computers and Electronics in Agriculture*, Elsevier, 20 May 2022.

Figures and Figure Captions

$$mAP = \frac{1}{k} \sum_i^k AP_i$$

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 * precision * recall}{precision + recall}$$

mAP: **0.8734**
Precision: **0.79**
Recall: **0.88**
F₁-score: **0.84**

Figure 1. YOLOv4-tiny performance metrics. Formulas and values of mAP, precision, recall, and F₁-score.

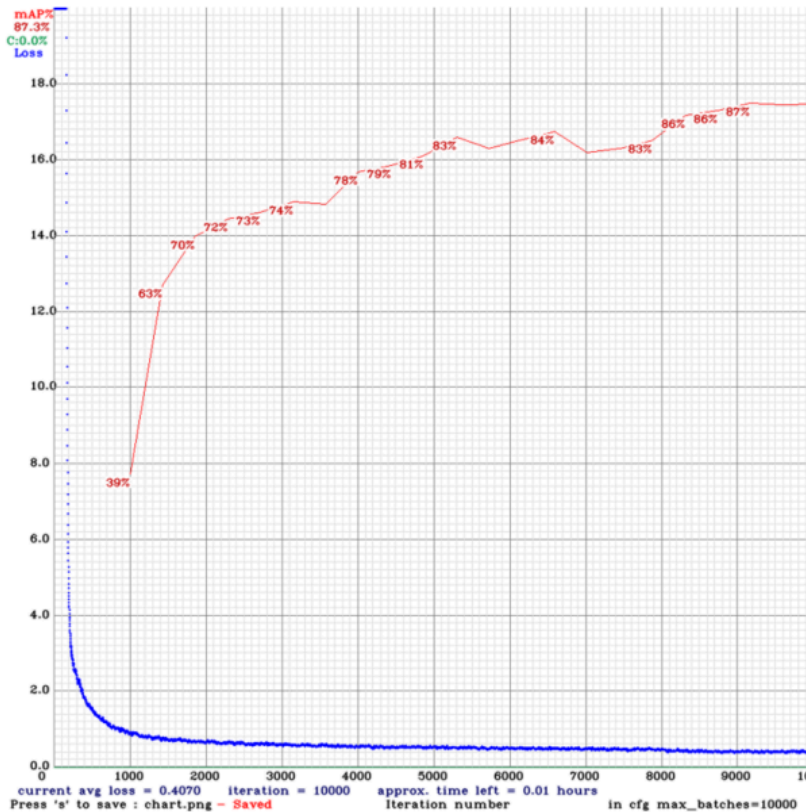


Figure 2. Training loss and mAP graph of the YOLOv4-tiny model. The blue line represents loss, and the red line represents mAP (10,000 iterations). These metrics were plotted on matplotlib and saved after training YOLOv4-tiny.

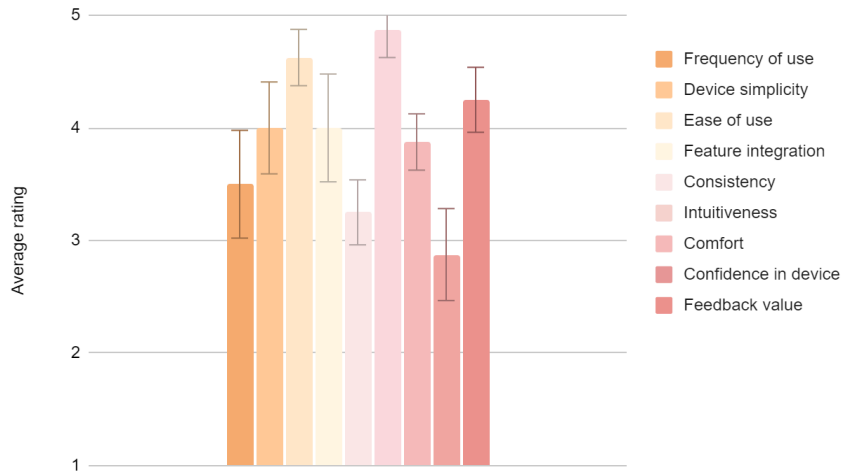


Figure 3. Average user ratings of the device across surveyed parameters. Bar graph showing mean $\pm$ SE ($n=8$). Users tested the apparatus at crosswalks and filled out a questionnaire based on the System Usability Scale, evaluating each variable on a scale of 1-5.



Figure 4. Annotated frame from the custom dataset (left) and real-time model predictions on a test image (right).

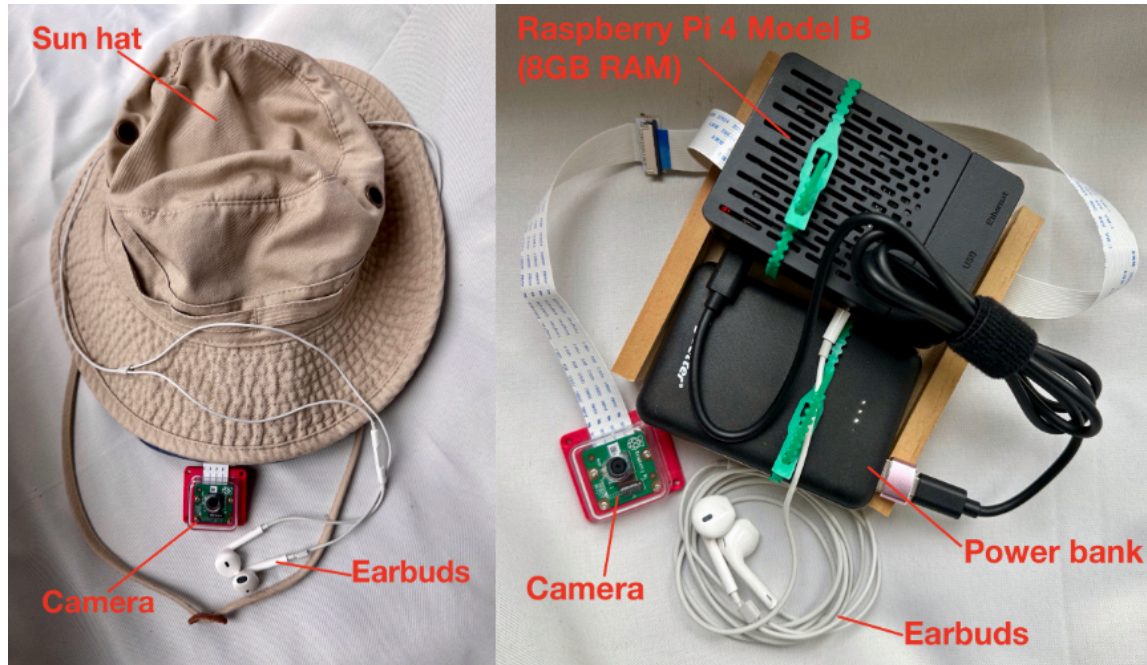


Figure 5. Photographs of the wearable apparatus, with (left) and without the hat (right).

Tables with Captions

Survey Prompts and Responses (Scale of 1-5)									
Prompt	#1	#2	#3	#4	#5	#6	#7	#8	avg
I think that I could use this device frequently.	2	2	4	3	4	5	5	3	3.5
I found the device to be simple.	4	3	5	4	3	4	5	4	4.0
I thought the device was easy to use.	4	5	5	5	4	4	5	5	4.6
I found the various functions in this device were well integrated.	5	3	5	4	2	5	4	4	4.0
I thought there was consistency in the device.	2	2	3	3	4	5	4	3	3.3
I would imagine that most people would learn to use this device very quickly.	5	5	5	4	5	5	5	5	4.9
I do not think the device was very cumbersome to use.	4	3	4	4	4	4	4	4	3.9
I felt very confident using the device .	2	1	3	2	4	4	4	3	2.9
I thought the device returned valuable information.	4	4	5	5	3	5	4	4	4.3

Table 1. Prompts and responses in the survey given to participants when testing the device (n=8). avg = mean of survey responses for a given prompt.

Class	# of Annotations	mAP
car	1,000	0.9286
crosswalk	1,810	0.9676
crosswalklightblank	65	
crosswalklightcountdown	184	
crosswalklightgo	730	0.8371
crosswalklightstop	1,672	0.7925
sidewalkend	1,688	0.8372

Table 2. Details about classes of the custom dataset. # of Annotations = quantity of annotations in given class; mAP = mean average precision of given class.

Appendices

Public Roboflow datasets used to train the model:

<https://universe.roboflow.com/sutees27/crosswalk-cluhl>

<https://universe.roboflow.com/walkassist/walk-assist>

<https://universe.roboflow.com/alexander-esenwein/crosswalk-aup9f>

- This should be part of the appendix because links, which cannot be included in the references, are the most specific way to access these datasets.